Do Now · retrieval

Processor and memory synoptic challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an ADC used for?

Q2

Which register holds the current instruction?

Q3

What is the main limitation of two cores on wholly dependent steps?

Q4

A full adder receives 1,1 and carry-in 0. Give sum and carry-out and explain.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Fetch the instruction — PC identifies address 72, which MAR sends to memory. MDR transfers its contents to CIR.

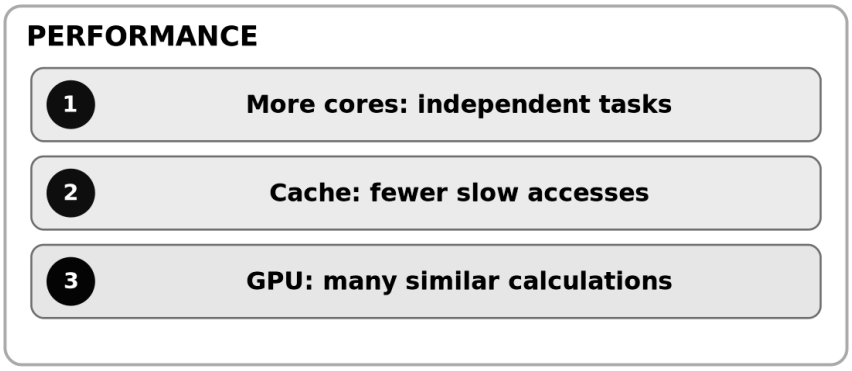
Core idea 2

Differentiate performance improvements — Cores can execute independent work; cache helps when needed data/instructions are present.

Core idea 3

Explain limits — Branching, shared memory bandwidth, serial dependencies and software constrain real speedup.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

PC — Program Counter; next instruction address.

MAR — Memory Address Register.

MDR — Memory Data Register.

cache — Fast nearby memory.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

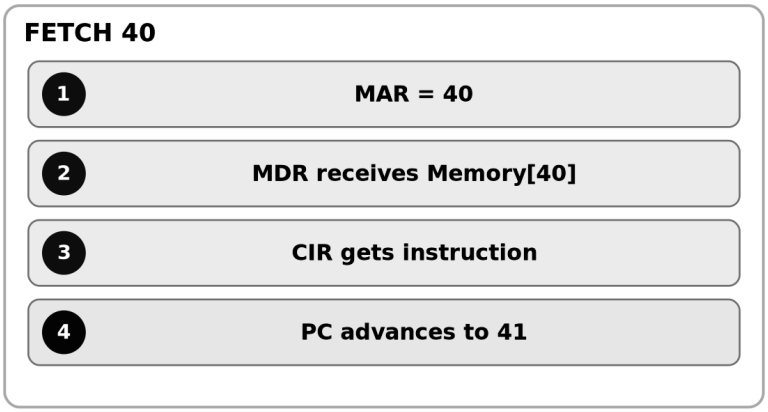
Starting PC=40, trace MAR, CIR and PC after a sequential fetch.

Follow the worked method

1. Predict — Copy PC=40 into MAR to select the instruction location.

2. Explain — Instruction contents travel via MDR to CIR, while sequential PC advances to 41.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Each available core receives a separate independent job.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State what PC, MAR, MDR, CIR and ACC contain or do.

Question 2

Explain how cache and extra cores could help video rendering, and why speedup may be less than expected.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare a multicore CPU and GPU for an interactive logic controller and a million independent pixels.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Doubling CPU cores always halves runtime.

Use key vocabulary from memory

1. Explain PC in your own words.

2. Explain MAR in your own words.

3. Explain MDR in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.